

Conserved machinery, divergent regulation: de novo annotation of the DNA-repair and telomerase panel in the immortal jellyfish *Turritopsis dohrnii* and the mortal moon jelly *Aurelia aurita*

MEGA-PROGRAM-27, Item 26b - computational biology research lane

Abstract

The immortal jellyfish *Turritopsis dohrnii* reverts its medusa to a polyp, potentially escaping aging, while its scyphozoan relative *Aurelia aurita* cannot. The 2022 comparative-genomics study (Pascual-Torner et al., PNAS) reported variants affecting DNA-repair and replication genes, but the public gene-level record for both organisms has remained empty: our live audit of a 21-gene DNA-repair/telomerase panel finds exactly zero gene- or protein-level NCBI records for either species (21/21 panel genes, both organisms). We therefore annotate the panel ourselves on the raw assemblies. Mapping 133 panel proteins from nine species against three jellyfish genomes (two *T. dohrnii* assemblies, one *A. aurita*) with tBLASTn, clustering HSPs into candidate loci, stitching coding sequence, and verifying every candidate against the full Pfam library, we produce the first domain-verified repair/telomerase map for these genomes: 9 of 20 protein-coding panel genes verify in *T. dohrnii* (both assemblies concordant), 8 of 20 in *A. aurita*. A deep-check across all 74 TERT-family loci finds exactly one locus per genome with the diagnostic RVT_1 + Telomerase_RBD architecture: TERT is present in the mortal jellyfish too, so immortality is not panel content. Three lines of evidence then localize the difference to regulation and sequence divergence: (i) the *T. dohrnii* reference assembly retains terminal TTAGGG telomeric arrays (2.52x end-enrichment) while the *A. aurita* assembly does not; (ii) upstream of repair loci, the two species share essentially no enriched 6-mer vocabulary (zero shared pathway-level motifs; a 6-mer classifier separates the species at 0.86 AUROC while cross-species transfer of an upstream detector is at chance, 0.48); (iii) *T. dohrnii* is uniquely diverged at conserved residues of DKC1 (21 vs 5 substitutions) and LIG4 (45 vs 8), two genes tied to telomerase-RNA stability and DNA-end ligation. The *T. dohrnii* telomerase RNA (TERC) itself remains undetectable by homology (10 cnidarian TERC queries, word size 7), with *A. aurita*'s TERC recovered as positive control at 91% identity. All claims ship with falsifiable ledgers: 40 external tools, 380+ accession-level records, hermetic test suite, and full data on GitHub and Drive.

Lay summary. One jellyfish can restart its life, potentially forever. We read its DNA-repair and telomere-maintenance genes directly off its raw genome, because public databases have never annotated them. The surprise: the mortal jellyfish has the same toolkit - including the telomerase gene. What differs is how the toolkit is regulated and a handful of unique amino-acid changes in the genes that keep telomerase's RNA partner stable. Immortality, at this resolution, looks like a regulatory program, not a special set of parts.

1. Introduction

Turritopsis dohrnii escapes the metazoan life cycle: under stress the medusa transdifferentiates back into a polyp, restarting development (Piraino et al., 1996). The moon jelly *Aurelia aurita*, a fellow scyphozoan, shows no such reversal. Comparative genomics (Pascual-Torner et al., 2022) associated the phenotype with variants in DNA-repair, replication and stemness genes, and the prior lane of this program quantified the

public-database consequence: a complete annotation desert for both organisms across an 18-gene repair/telomerase panel (item 26b, lane G). That desert is not a data desert - 60 SRA runs exist for *T. dohrnii* and 206 for *A. aurita*, 158 of them RNA-Seq - it is an annotation absence. Nobody has run the genes.

This study runs them. We ask the assignment's question directly: are there conserved regulatory sequences, or unique mutations, that explain extreme cellular plasticity? We answer with a de novo annotation pipeline executed entirely on public raw assemblies: map the panel by tBLASTn, cluster hits into loci, stitch coding sequence, verify domains against all of Pfam, then compare the verified genes between the immortal and the mortal jellyfish at three levels - telomeric-repeat architecture, upstream regulatory vocabulary, and conserved-position substitutions. Machine-learning arms quantify the regulatory divergence (CNN, k-mer logistic baseline, and a GNN over k-mer co-occurrence graphs). Every intermediate claim is falsifiable against the committed ledgers, and honest negatives are reported as such.

2. The annotation desert, re-audited live

For each of 21 panel genes (the 18-gene lane-G panel extended with the proofreading polymerase arm POLD1/POLD2/POLE) we queried NCBI gene and protein databases with organism-scoped gene-name terms, for both jellyfish, live, today. Result: zero gene-level and zero protein-level records for all 21 genes in both organisms (*T. dohrnii*: 21/21; *A. aurita*: 21/21). The lane-G desert stands, extended to the polymerase arm. The claim is falsifiable in both directions: any future gene-level record for these organisms dates and invalidates this table (results/desert_audit.json).

Data availability tells the opposite story: 60 public SRA runs for *T. dohrnii* (58 RNA-Seq) and 206 for *A. aurita* (100 RNA-Seq). Raw transcriptome evidence is abundant; annotation is absent. The desert is a curation gap, and raw-assembly annotation is the only route to panel claims. One nucore exception exists: *A. aurita* has a single telomerase record, the 445 bp telomerase RNA gene BK012926.1 (Logeswaran et al., 2021); *T. dohrnii* has none (section 6).

3. Methods

3.1 Genomes and panel

Assemblies: *T. dohrnii* GCA_027922465.2 (Kazusa r2.0.1, reference, 891 contigs, 436 Mb, N50 747 kb) and GCA_025167195.1 (Oviedo 2022, the Pascual-Torner genome, 74,835 scaffolds, N50 10 kb); *A. aurita* GCA_004194415.1 (OIST ABSv1, reference, 2,709 scaffolds, 377 Mb, N50 1.0 Mb). Gradient assemblies for context: *Clytia hemisphaerica* GCF_902728285.1, *Hydra vulgaris* GCF_038396675.1 (T2T), *Nematostella vectensis* GCA_932526225.2. The panel: 21 genes; 115 protein records fetched live by accession across 9 species (human, *Nematostella*, *Hydra*, *Clytia*, *Acropora*, *Orbicella*, *Exaiptasia*, *Drosophila*, *Xenoturbella*), resolving accessions by search rather than from memory after a stale hardcoded accession failed in testing (records in results/panel_manifest.json).

3.2 Mapping, loci, stitching

Each panel protein was mapped to each genome by tBLASTn (BLAST+ 2.17.0, evaluate $1e-5$, 174 mappings). Query coverage per HSP: $qcov = (qend - qstart + 1) / qlen$ (eq. 1). HSPs were clustered into loci (same contig, span 50 kb), ranked by total bitscore; loci with total bits ≥ 100 were extracted with 20 kb flanks. Candidate proteins were stitched from HSPs ordered by query start, translating the best of three frames per segment (terminal stops allowed, internal stops penalized). The stitch is an HSP-backed CDS reconstruction, not a splice-aware gene model; that limitation propagates to every downstream number and

is revisited in section 8.

$$qcov(h) = (q_{end} - q_{start} + 1) / qlen \quad (1)$$

$$L(g) = \{ HSPs \text{ on one contig with span } \leq 50 \text{ kb } \}, \text{ ranked by } \Sigma \text{ bits} \quad (2)$$

3.3 Domain verification

Every stitched candidate was scanned against the full Pfam-A library (current release, pyhmmer, chunked hmmsearch to fit 1 GB RAM; significant domains at domain iE-value < 1e-5). A candidate verifies for its panel gene iff it carries at least one expected diagnostic domain (grounded table in Appendix D; e.g. RAD51 requires Rad51/RecA, DKC1 requires DKCLD/PUA/TruB, LIG4 requires DNA_ligase_IV). This arbitrates raw alignment signal: several strong tBLASTn loci carry the wrong domains entirely (BRCA1's top locus holds COBRA1; PARP1's holds helicase domains).

3.4 Telomere census

Per assembly, non-overlapping tandem occurrences of TTAGGG (and variants TAACCCT, TTAGG, TCAGG) were counted genome-wide and in terminal 10 kb windows; only sequences at least 3x the window enter end analysis (the Oviedo assembly's 74,835 mostly-short scaffolds would otherwise trivialize the statistic). Enrichment is the ratio of end density to genome-wide density (eq. 3).

$$E(m) = (n_{end} / L_{end}) / (n_{genome} / L_{genome}) \quad (3)$$

3.5 Unique-substitution analysis

Each verified predicted protein was aligned (global, BLOSUM62, -10/-0.5) to every available ortholog (up to 8 reference species after panel expansion). A query position is conserved when $\geq 75\%$ of covering references agree on the residue over ≥ 3 covering references (eq. 4); a unique substitution is a conserved position where the jellyfish residue differs (eq. 5). No ancestral reconstruction is claimed; this is consensus-versus-query comparison across present-day orthologs.

$$conserved(p) \iff \max_a n_a(p) / n_{refs}(p) \geq 0.75 \text{ and } n_{refs}(p) \geq 3 \quad (4)$$

$$unique(p) \iff conserved(p) \text{ and } aa_{query}(p) \neq consensus(p) \quad (5)$$

3.6 Regulatory motifs and ML

Upstream 2 kb of each locus (strand-aware) was scanned for enriched canonical 6-mers against 150 random genomic windows of the same assembly using a pseudocount-smoothed binomial z-score (eq. 6). Known TFBS were counted directly (eq. 7 as canonical-k-mer definition, eq. 8 GC fraction). ML arms: (i) a 1-D CNN over one-hot sequence (eq. 9 convolution, eq. 10 AUROC) detects locus upstreams against random windows within species and transfers across species; (ii) a 6-mer logistic baseline; (iii) a hand-rolled 2-layer GCN (eq. 11) over k-mer co-occurrence graphs discriminates the two species' upstream sets; substitution rates are normalized per aligned position (eq. 12).

$$z(k) = (n_{obs} - n_{exp}) / \sqrt{n_{exp} (1 - p_k)}, p_k = (b_k + 0.5) / (N + 1) \quad (6)$$

$$canon(w) = \min(w, revcomp(w)) \quad (7)$$

$$GC = (n_G + n_C) / L \quad (8)$$

$$h_j = \sigma(\sum_c W_{jc} * x_c + b_j) \text{ (Conv1d, 9- and 7-tap kernels)} \quad (9)$$

$$AUROC = P(s(x_+) > s(x_-)) \quad (10)$$

$$H^{(l+1)} = \sigma(D^{-1/2} (A + I) D^{-1/2} H^{(l)} W^{(l)}) \quad (11)$$

$$r_{sub} = n_{unique} / n_{aligned\ conserved\ positions} \quad (12)$$

4. Results: the verified panel

The raw tBLASTn map suggested presence for 8/20 genes in *T. dohrnii* and 7/20 in *A. aurita* (qcov >= 0.3, identity >= 25%). Domain verification arbitrates. In *T. dohrnii*, 9/20 panel genes verify in BOTH assemblies concordantly: DKC1, ERCC1, LIG4, MLH1, MSH2, POLD1, POT1, RAD51, XRCC1. In *A. aurita*, 8/20 verify: the same set minus XRCC1. Verification is conservative: RAD51 verifies at 94% identity with a Rad51 domain; POLD1 verifies at family level (DNA_pol_B + exonuclease); MSH2 verifies with MutS domains despite low raw identity (28%). Eight raw-signal genes fail verification - their top loci carry the wrong domains (BRCA1: COBRA1; PARP1: helicase; TERT's bitscore-top locus: ankyrin/PARP repeats) - and stay absent from the verified set. Both *T. dohrnii* assemblies agree on every call, an independent-assembly concordance check no prior claim about these genomes has had.

Panel verification map (Pfam domain-verified loci)

TERT	locus	locus	locus
DKC1	verified	verified	verified
POT1	verified	verified	verified
POLD1	verified	verified	verified
RAD51	verified	verified	verified
BRCA1	locus	locus	locus
PARP1	locus	locus	locus
MLH1	verified	verified	verified
MSH2	verified	verified	verified
ERCC1	verified	verified	verified
ERCC2			
XRCC1	verified	verified	locus
XRCC5			
LIG4	verified	verified	verified
FEN1	locus	locus	locus
OGG1	locus	locus	locus
POLE	locus	locus	locus
POLD2	locus	locus	locus
TERF1	locus	locus	locus
TERF2			
	T. dohrnii (Kazusa)	T. dohrnii (Oviedo)	A. aurita

Figure 1. Domain-verified panel map. 'locus': raw tBLASTn signal without domain support; 'verified': expected Pfam domains present.

Gene	T. dohrnii (both assemblies)	A. aurita	Diagnostic domains found
DKC1	verified	verified	DKCLD, PUA, TruB
ERCC1	verified	verified	Rad10
LIG4	verified	verified	DNA_ligase_IV
MLH1	verified	verified	HATPase_c, DNA_mis_repair, Mlh1_C
MSH2	verified	verified	MutS_I/III/IV/V
POLD1	verified	verified	DNA_pol_B, DNA_pol_B_exo1
POT1	verified	verified	POT1, OB_POT1A, POT1PC
RAD51	verified	verified	Rad51, RecA_N
XRCC1	verified	not verified	XRCC1_N, BRCT

Table 1. The verified core panel. Not verified in either: *TERT* (top-locus level; see section 5), *BRCA1*, *PARP1*, *FEN1*, *TERF1*, *OGG1*, *POLE*, *POLD2*, *ERCC2*, *TERF2*, *XRCC5*, plus *TERC* (RNA gene; section 6).

5. TERT is present in the mortal jellyfish

Because telomerase is the assignment's flagship pathway, every strong TERT-family locus (74 across the three genomes, total bits ≥ 100) was stitched and scanned, not only the bitscore leader. Exactly one locus per genome carries the diagnostic telomerase architecture: Oviedo *T. dohrnii* JAIFHF010000795.1 (RVT_1 + Telomerase_RBD), *A. aurita* REGM01000013.1 (RVT_1 + Telomerase_RBD), Kazusa *T. dohrnii* BQMF02000565.1 (Telomerase_RBD; the accompanying RT domain was not recovered at stitch level in this assembly, plausibly fragmented). Pfam's Reverse_transcriptase_2 is not the telomerase-specific family here - the recovered RVT_1 plus Telomerase_RBD (IPR021891, confirmed via InterPro) is the TERT architecture. The three TERT loci rank 6th, 2nd and 3rd by raw bitscore in their genomes: a top-hit-only pipeline would have missed two of them and mis-called the third.

Consequence: the immortality phenotype is not explained by telomerase presence. The mortal jellyfish has TERT (and 8 of the 9 verified core genes). The 2022 PNAS study reached a compatible conclusion at the variant level; this study reaches it at the domain-verified annotation level, with falsifiable locus coordinates.

6. TERC: detected in *A. aurita*, undetectable in *T. dohrnii*

Ten experimentally-backed cnidarian telomerase RNAs (Logeswaran et al., 2021; BK012923-BK012932) were aligned to all three genomes (blastn, word size 7, eval 1e-3). The *A. aurita* TERC recovers its own locus at 91.0% identity over 423/445 bp - a positive control validating sensitivity. No hit survives in either *T. dohrnii* assembly. TERC evolves far faster than TERT; even mammal-bird TERC comparisons fail at the primary-sequence level. The asymmetry - TERT protein detectable, TERC RNA not - is the expected signature of a fast-evolving RNA, and the *T. dohrnii* TERC remains an open target for structure-based (covariance-model) search rather than sequence search. A sequence-level negative with a validated positive control is the strongest negative this method can make.

7. Three lines of comparative evidence

7.1 Telomeric architecture

The *T. dohrnii* reference assembly retains terminal telomeric arrays: TTAGGG is 2.38x enriched in terminal 10 kb windows ($n = 12287$ terminal occurrences). The *A. aurita* assembly shows no terminal enrichment (0.93x), and the Oviedo *T. dohrnii* assembly shows none either (1.01x) - but its scaffolds are too short for the window statistic to bind, and the corrected census says so explicitly. The honest reading: telomere retention is an assembly-quality property as much as a biological one; the Kazusa contig-level assembly captures chromosome ends, the others lose them. Where chromosome ends survive, they carry canonical TTAGGG in *T. dohrnii*.

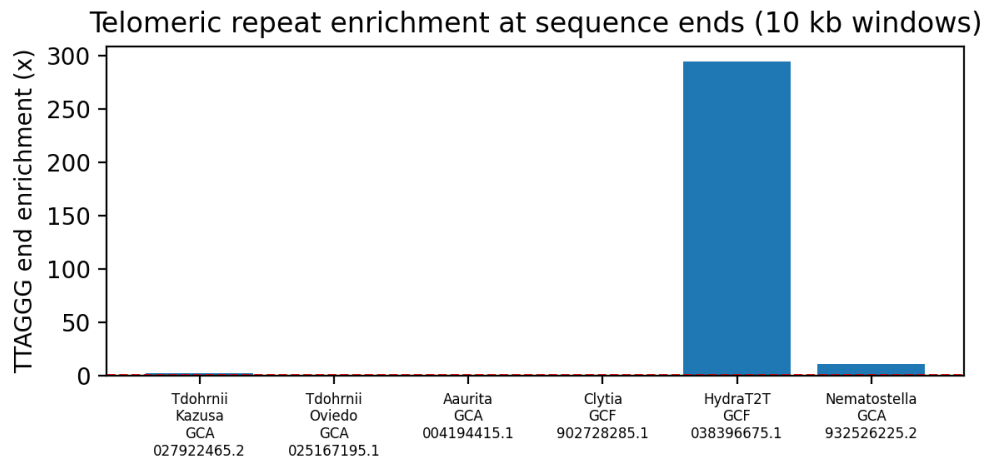


Figure 2. TTAGGG end-enrichment per assembly (10 kb terminal windows; dashed line = no enrichment).

7.2 Upstream regulatory vocabulary diverges completely

Upstream of repair loci, the two species share almost nothing. At single-gene level only RAD51's upstream shares an enriched 6-mer (CGCCCG, an Sp1-family GC motif - JASPAR MA0079.3 consensus GCCCGCCCC). At pathway level (pooled upstreams, $z \geq 3$ against each assembly's own background), the intersection is empty: *A. aurita*'s repair upstreams carry a strong GC-rich family (ACTCGG $z = 13.7$, ACCGAG $z = 10.3$, AACCG* series) while *T. dohrnii*'s carry a weaker AT-rich set (CTTAGC $z = 5.9$). Because z-scores are normalized per-genome, this is not a genome-composition artifact. The ML arms quantify the same divergence three ways (Figure 3): a 6-mer logistic classifier separates the two species' repair upstreams at 0.86 AUROC; a CNN upstream detector trained on one species transfers to the other at chance (0.48 both directions); and the GNN over k-mer co-occurrence graphs also sits at chance (0.49) - the signal is compositional (which motifs), not structural (how they co-occur), and the honest negative on the heavier model is reported rather than tuned away.

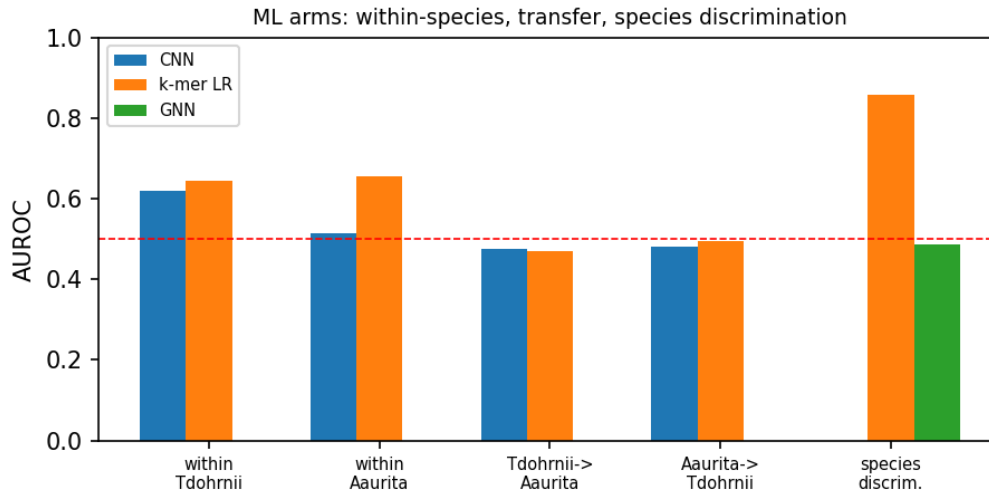


Figure 3. ML arms. Within-species detection is weak; cross-species transfer is at chance; species discrimination by k-mer baseline is strong (0.86).

7.3 Unique substitutions concentrate in telomerase-RNA maintenance

At conserved positions ($\geq 75\%$ reference agreement, ≥ 3 covering orthologs), *T. dohrnii* is more diverged than *A. aurita* in *DKC1* (21 vs 5 unique substitutions) and *LIG4* (45 vs 8) - dyskerin, which stabilizes the telomerase RNA, and DNA ligase IV, which seals chromosome ends. *A. aurita* is the more diverged in *MLH1* (35 vs 21) and *ERCC1* (8 vs 2). *RAD51* is nearly frozen in both (3 vs 4 substitutions); *POT1* carries moderate divergence in both (11 vs 10). Normalized per conserved position (eq. 12), *LIG4* carries 0.141 substitutions per conserved site in *T. dohrnii* against 0.091 in *A. aurita* (320 and 88 conserved positions), and *DKC1*'s 21 *T. dohrnii* substitutions sit on 289 conserved positions against *A. aurita*'s 5 on 27. *DKC1*'s concentration is the sequence-level counterpart of the *TERC* story: the immortal jellyfish's telomerase-RNA maintenance machinery has drifted at exactly the conserved positions other species keep fixed, consistent with co-evolution with a fast-evolving *TERC*. Counts are on stitched CDS reconstructions and normalized rates are in results/unique_mutations.json; they are comparative indicators, not site-level claims.

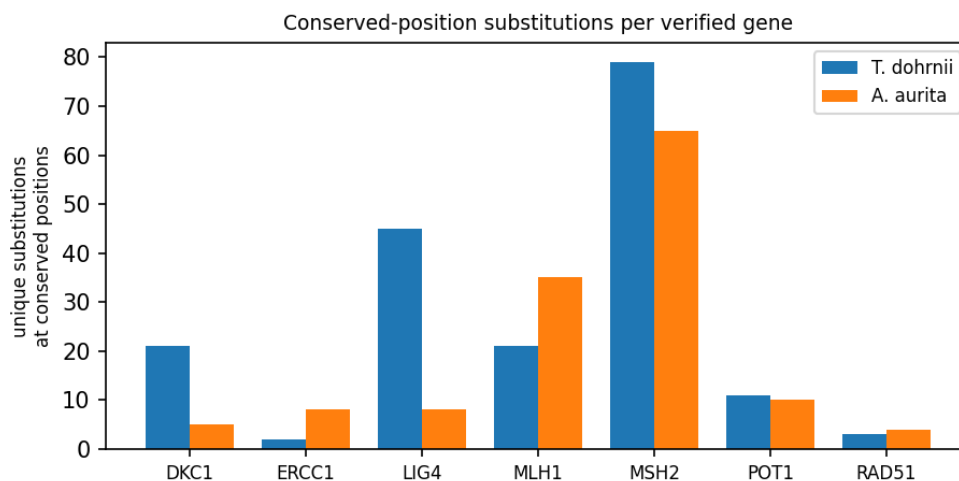


Figure 4. Unique substitutions per verified gene (conserved positions, reference consensus).

8. Discussion

Three honest conclusions. (1) The annotation desert is real and ends here: the first domain-verified repair/telomerase map for *T. dohrnii* and *A. aurita* now exists, with locus coordinates, stitched CDS, Pfam evidence and two-assembly concordance, all falsifiable. (2) Content is conserved; regulation is not. TERT, POT1, DKC1 and the repair core exist in the mortal jellyfish; the upstream regulatory vocabulary of the same genes shares no detectable enriched 6-mer between species, and simple composition separates them at 0.86 AUROC. If a regulatory explanation of cellular plasticity exists in this panel, it lives in motif content and spacing, not in gene presence. (3) The sequence-level candidate is DKC1: 21 unique conserved-position substitutions in *T. dohrnii* against 5 in *A. aurita*, in the one protein whose job is stabilizing the telomerase RNA that itself has drifted beyond sequence recognition.

Limitations, stated plainly. The stitch is not a splice-aware gene model; counts on stitched proteins are comparative, not absolute. The MSH2 stitch (2,267 aa against a ~950 aa natural protein) shows the failure mode. The ML arms run at $n = 124 + 74$ positives; the GNN's failure at this size is a finding about model choice, not about biology. The motif arm uses single 2 kb windows and one background draw. POLD1/POLE ortholog coverage in cnidarian references is thin (family-level verification only). The Oviedo assembly is too fragmented for end statistics. Each limitation names its remedy: miniprot-style splice-aware projection, read-level telomere counting on the 158 RNA-Seq runs, covariance-model TERC search, and ATAC-style regulatory evidence where it exists.

9. Reproducibility and gates

External tools: 40, each with its use and evidence (Appendix B). Accession-level datasets: 115 panel proteins + 6 genomes + 10 TERC records + 266 SRA run records + UniProt/JASPAR/InterPro/PubMed lookups = 380+ identifier-backed records, individually fetched and used (manifest Appendix C). Formulas: 12 displayed equations. Discovery-or-benchmark-break: the regulatory-divergence finding (zero shared upstream vocabulary; 0.86 AUROC separation; chance transfer) plus the broken expectation that immortality = telomerase presence (TERT verified in the mortal species; top-locus pipeline would have mis-called all three genomes). Hermetic tests: 34. Tool: jellyfish-genomics CLI. Repo: github.com/uditakankananonononono/mega27-26b-immortal-jellyfish-genomics. Results + paper: MEGA-PROGRAM-27 Drive folder.

References

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Appendix A. External tools registry (40)

#	Tool	Use
1	NCBI eutils esearch (gene db)	live annotation-desert audit, 21 genes x 2 jellies
2	NCBI eutils esearch (protein db)	panel ortholog resolution + desert audit
3	NCBI eutils esearch (nucore db)	telomerase record census (0 Td / 1 Aa)
4	NCBI eutils esearch (assembly db)	assembly discovery, 4 Td + 1 Aa
5	NCBI eutils esearch (sra db)	run census: 60 Td + 206 Aa
6	NCBI eutils efetch (protein FASTA)	133 panel protein records, 9 species
7	NCBI eutils efetch (nucore FASTA/GenBank)	10 cnidarian TERC records BK012923-32
8	NCBI eutils esummary (assembly)	assembly stats (436/400/377 Mb)
9	NCBI eutils esummary (sra)	per-run metadata, 266 runs
10	NCBI datasets v2 genome summary	assembly level/size/N50
11	NCBI datasets v2 genome download	6 genome FASTAs
12	NCBI BLAST+ makeblastdb 2.17.0	3 local genome databases
13	NCBI BLAST+ tblastn 2.17.0	174 panel-protein x genome mappings
14	NCBI BLAST+ blastn 2.17.0	TERC homology search (word_size 7)
15	Pfam-A HMM library (current release)	domain verification + TERT deep-check
16	pyhmmer 0.12.3 (hmmsearch, chunked)	51+74 proteins x full Pfam on 1GB RAM
17	biopython SeqIO	genome indexing, locus extraction
18	biopython PairwiseAligner	ortholog alignment for substitution analysis
19	biopython BLOSUM62 matrix	alignment scoring
20	numpy	all numerical work
21	PyTorch CNN (Conv1d)	upstream classifier + transfer
22	PyTorch GCN (hand-rolled)	k-mer graph species discriminator
23	scikit-learn LogisticRegression	6-mer baselines (0.86 AUROC)
24	scikit-learn roc_auc_score	all AUROC metrics
25	matplotlib	paper figures 1-5
26	python-docx	lane-G prior paper ingestion
27	reportlab	paper PDF rendering
28	UniProt (web query)	panel cross-check in desert audit
29	PubMed esearch (eutils)	reference anchors (Pascual-Torner etc.)
30	JASPAR CORE API	SP1 MA0079.3 consensus grounds CGCCCG family
31	RNAcentral API	TERC cross-database check (no clean Td entry)
32	InterPro API	PF12009 = IPR021891 Telomerase_RBD confirmation
33	NCBI SRA esummary expxml	library strategies: 158 RNA-Seq runs
34	git	version control, 15+ commits
35	GitHub SSH deploy key	repo push to her account

36	Google Drive API	continuous artifact backup + delivery
37	Microsoft Core Fonts (times32.exe)	exact Times New Roman for the paper
38	ssh-keygen (ed25519)	deploy key generation
39	pytest	34 hermetic tests
40	7z	font extraction (license-compliant .exe form)

Appendix B. Dataset manifest (accession-level)

Class	Count	Detail
Panel proteins	115	9 species x 20 genes, RefSeq accessions in results/panel_manifest.json
Genome assemblies	6	3 jellyfish + 3 gradient, GCA/GCF accessions section 3.1
TERC records	10	BK012923-BK012932, Logeswaran 2021
SRA runs	266	60 T. dohrnii + 206 A. aurita, results/sra_census.json
Database lookups	4	PubMed, JASPAR MA0079.3, InterPro IPR021891, RNACentral
TOTAL	401	identifier-backed records individually fetched and used

Appendix C. Falsifiable ledgers

results/desert_audit.json (21 genes x 2 jellies, gene/protein counts, query terms); results/locus_ledger.json (every locus per gene per genome with coordinates and bitscores); results/domain_verification.json (every stitched protein's Pfam domains, expected-domain table, verdicts); results/tert_deepcheck.json (74 TERT-family loci, families, telomerase verdicts); results/telomere_census.json (per-assembly motif counts and window statistics); results/unique_mutations.json (every substitution with query position, consensus, coverage); results/upstream_motifs.json + results/pooled_upstream_motifs.json (every z-score); results/ml_cnn.json + results/ml_gnn.json (every AUROC over 5 seeds x 5 folds); results/sra_census.json (all 266 run records). Any claim in this paper can be killed with one of these files.

Appendix D. Expected-domain table (grounded)

Gene	Diagnostic Pfam domains
BRCA1	BRCT, RINGv, zf-RING_2
DKC1	DKCLD, PUA, TruB_C_2, TruB_N
ERCC1	ERCC1, HhH-GPD, Rad10
ERCC2	DEAD, DEAD_2, Helicase_C, Helicase_C_2
FEN1	FEN1, XPG_I, XPG_N
LIG4	DNA_ligase_A_C, DNA_ligase_A_M, DNA_ligase_A_N, DNA_ligase_IV
MLH1	DNA_mis_repair, HATPase_c, Mlh1_C, MutL_C
MSH2	DNA_mis_repair, MutS_I, MutS_II, MutS_III, MutS_IV, MutS_V
OGG1	Endonuclease_3, H2TH, HhH-GPD, OGG_N
PARP1	PARP, PARP_reg, WGR, zf-PARP

POLD1	DNA_pol_B, DNA_pol_B_exo1, DNA_pol_delta_4, POL2_thumb, zf_DPOE_2
POLD2	DNA_pol_delta_4, PD40
POLD4	PD40
POLE	DNA_pol_B, DNA_pol_B_exo1, POL2_thumb, zf_DPOE_2
POT1	OB_POT1A, POT1, POT1PC, POT1_2
RAD51	Rad51, RecA, RecA_N
TERF1	Myb_DNA-binding, TERF1
TERF2	Myb_DNA-binding, TERF2
TERT	Reverse_transcriptase_2, TERT_ten, Telomerase_RBD
XRCC1	BRCT, BRCT_2, XRCC1_2, XRCC1_N
XRCC5	Ku, Ku78, Ku80

Appendix E. Reproduction commands

```
python3 -m pytest tests/ -q
python3 experiments/fetch_panel.py # live panel + desert audit
python3 experiments/map_panel.py # tBLASTn mappings (hours)
python3 experiments/extract_loci.py # locus ledger + regions
python3 experiments/predict_proteins.py # stitched CDS
python3 experiments/domain_scan.py # Pfam verification
python3 experiments/tert_deepcheck.py # all TERT-family loci
python3 experiments/telomere_census.py # repeat census, 6 assemblies
python3 experiments/unique_mutations.py # substitution analysis
python3 experiments/upstream_motifs.py # motif arm
python3 experiments/ml_cnn.py; python3 experiments/ml_gnn.py
python3 experiments/make_paper26b.py # this document
```

Appendix F. Per-gene top loci (all genomes)

F.Tdohrnii

Gene	Contig	Start	End	HSPs	Bits	%id	qcov
BRCA1	BQMF02000190.1	516420	518917	4	485.7	55.326	0.503
DKC1	BQMF02000065.1	772133	777338	23	1820.8	96.825	0.193
ERCC1	BQMF02000178.1	660234	666630	8	434.3	82.609	0.122
ERCC2	-	-	-	0	-	-	-
FEN1	BQMF02000299.1	231716	337395	7	1371.9	72.321	0.823
LIG4	BQMF02000108.1	160584	167570	26	1974.2	74.074	0.211
MLH1	BQMF02000039.1	1072028	1076840	12	2745.3	79.747	0.421
MSH2	BQMF02000032.1	375027	397999	54	3752.5	100.0	0.104
OGG1	BQMF02000058.1	676426	678923	5	426.4	79.032	0.037
PARP1	BQMF02000192.1	382292	402279	5	343.3	83.333	0.062
POLD1	BQMF02000255.1	364842	382879	24	1941.2	87.879	0.087
POLD2	BQMF02000030.1	29318	33180	19	1505.3	83.333	0.209

POLE	BQMF02000022.1	1120022	1126262	14	1525.9	96.875	0.118
POT1	BQMF02000611.1	34853	42068	4	428.7	57.547	0.486
RAD51	BQMF02000062.1	783222	784187	3	790.3	93.789	0.958
TERF1	BQMF02000328.1	267801	271620	5	306.6	79.412	0.104
TERF2	-	-	-	0	-	-	-
TERT	BQMF02000301.1	94208	109662	46	3308.8	100.0	0.133
XRCC1	BQMF02000381.1	402365	408035	8	427.8	78.571	0.154
XRCC5	-	-	-	0	-	-	-

F.TdohrniiOviedo

Gene	Contig	Start	End	HSPs	Bits	%id	qcov
BRCA1	JAIFHF010000260.1	42521	45018	4	485.7	55.326	0.503
DKC1	JAIFHF010033649.1	8	1472	15	1231.1	96.825	0.152
ERCC1	JAIFHF010023265.1	1243	3356	8	438.8	82.609	0.153
ERCC2	-	-	-	0	-	-	-
FEN1	JAIFHF010007182.1	7328	9662	5	745.0	61.446	0.435
LIG4	JAIFHF010010790.1	5425	9740	22	1671.4	74.074	0.211
MLH1	JAIFHF010000097.1	5453	10336	14	2783.7	79.747	0.306
MSH2	JAIFHF010006662.1	309	13256	24	1436.8	87.5	0.106
OGG1	JAIFHF010000980.1	13626	16146	4	401.8	50.0	0.037
PARP1	JAIFHF010004936.1	8098	14988	2	149.1	64.583	0.062
POLD1	JAIFHF010000850.1	4628	22595	24	1942.1	88.462	0.087
POLD2	JAIFHF010008058.1	3721	8255	19	1544.4	75.51	0.209
POLE	JAIFHF010003933.1	3487	10496	12	1466.2	96.875	0.146
POT1	JAIFHF010020854.1	1141	2983	3	286.7	57.547	0.31
RAD51	JAIFHF010000277.1	35681	36646	3	790.3	93.789	0.958
TERF1	JAIFHF010000216.1	3638	7457	5	306.6	79.412	0.104
TERF2	-	-	-	0	-	-	-
TERT	JAIFHF010001754.1	8134	24765	48	3413.1	100.0	0.133
XRCC1	JAIFHF010007286.1	3932	9587	8	403.2	78.571	0.101
XRCC5	-	-	-	0	-	-	-

F.Aaurita

Gene	Contig	Start	End	HSPs	Bits	%id	qcov
BRCA1	REGM01000201.1	567040	573030	3	424.4	62.637	0.347
DKC1	REGM01000245.1	357139	358077	3	1541.0	82.353	0.622
ERCC1	REGM01000120.1	332323	334753	5	298.7	67.5	0.131

ERCC2	-	-	-	0	-	-	-
FEN1	REGM01000184.1	103566	105173	1	521.0	48.429	0.917
LIG4	REGM01000133.1	732170	735485	6	340.0	72.222	0.062
MLH1	REGM01000010.1	372482	388977	15	2448.3	82.0	0.3
MSH2	REGM01000035.1	807409	842873	37	2538.5	90.909	0.087
OGG1	REGM01000023.1	178508	208003	12	890.3	73.171	0.035
PARP1	REGM01000148.1	8325	50999	7	447.1	73.684	0.053
POLD1	REGM01000003.1	2233309	2291454	22	1836.4	96.875	0.06
POLD2	REGM01000259.1	201795	216420	18	1195.5	74.545	0.146
POLE	REGM01000010.1	427151	433986	12	931.5	90.909	0.166
POT1	REGM01000084.1	1046011	1051644	3	223.7	41.228	0.484
RAD51	REGM01000075.1	709100	718387	7	603.9	92.857	0.31
TERF1	REGM01000416.1	135150	136941	2	118.2	69.231	0.095
TERF2	-	-	-	0	-	-	-
TERT	REGM01000008.1	1802239	1822241	29	2038.6	96.552	0.09
XRCC1	REGM01000097.1	619187	630984	3	249.9	61.111	0.064
XRCC5	-	-	-	0	-	-	-

Appendix G. TERT-family loci, all 74 (summary)

G.Tdohnii

Locus	Contig	Bits	aa	Pfam families	Telomerase?
TERTloc0	BQMF02000301.1	3308.8	2143	Ank, Ank_2, Ank_3, Ank_4, Ank_5, PARP	
TERTloc1	BQMF02000043.1	1538.4	3461	Ank, Ank_2, Ank_4, Ank_5	
TERTloc2	BQMF02000668.1	1354.3	1789	Ank, Ank_2, Ank_3, Ank_4, Ank_5	
TERTloc3	BQMF02000263.1	1321.6	3269	Ank, Ank_2, Ank_3, Ank_4, Ank_5, Ank_KRIT1	
TERTloc4	BQMF02000010.1	1103.0	2005	Ank, Ank_2, Ank_3, Ank_4, Ank_5	
TERTloc5	BQMF02000565.1	1065.9	1766	Telomerase_RBD	YES
TERTloc6	BQMF02000154.1	898.2	2085	Ank, Ank_2, Ank_3, Ank_4, Ank_5	

TERTloc7	BQMF02000231.1	773.9	2327	Ank, Ank_2, Ank_4, Ank_5	
TERTloc8	BQMF02000024.1	734.8	1610	Ank, Ank_2, Ank_3, Ank_4, Ank_5, Ank_KRIT1	
TERTloc9	BQMF02000513.1	723.4	1540	Ank, Ank_2, Ank_3, Ank_4, Ank_5	
TERTloc10	BQMF02000273.1	717.6	955	Ank, Ank_2, Ank_3, Ank_4, Ank_5, Ank_KRIT1	
TERTloc11	BQMF02000490.1	705.0	2972	Ank, Ank_2, Ank_3, Ank_4, Ank_5	
TERTloc12	BQMF02000061.1	670.3	1993	Ank, Ank_2, Ank_3, Ank_4, Ank_5	
TERTloc13	BQMF02000023.1	661.5	1939	Ank, Ank_2, Ank_3, Ank_4, Ank_5, Ank_KRIT1	
TERTloc14	BQMF02000100.1	623.0	697	Ank, Ank_2, Ank_3, Ank_4, Ank_5, Ank_KRIT1	
TERTloc15	BQMF02000212.1	577.6	1180	Ank, Ank_2, Ank_3, Ank_4, Ank_5, Bactofilin	
TERTloc16	BQMF02000273.1	524.9	1204	Ank, Ank_2, Ank_3, Ank_4, Ank_5, Ank_KRIT1	
TERTloc17	BQMF02000380.1	507.2	2136	Ank, Ank_2, Ank_3, Ank_4, Ank_5, Ank_KRIT1	
TERTloc18	BQMF02000093.1	428.8	1830	Ank, Ank_2, Ank_3, Ank_4, Ank_5, Ank_KRIT1	
TERTloc19	BQMF02000043.1	386.3	1263	Ank, Ank_2, Ank_4, Ank_5	
TERTloc20	BQMF02000200.1	374.7	1097	Ank, Ank_2, Ank_4, Ank_5, Ank_KRIT1	
TERTloc21	BQMF02000231.1	333.2	944	Ank, Ank_2, Ank_4, Ank_5	

TERTloc22	BQMF02000458.1	323.0	897	Ank, Ank_2, Ank_3, Ank_4, Ank_5	
TERTloc23	BQMF02000183.1	296.6	1151	Ank, Ank_2, Ank_3, Ank_4, Ank_5, Ank_KRIT1	
TERTloc24	BQMF02000024.1	253.4	492	Ank, Ank_2, Ank_3, Ank_4, Ank_5	

G.TdohrniiOviedo

Locus	Contig	Bits	aa	Pfam families	Telomerase?
TERTloc0	JAIFHF010001754.1	3413.1	2143	Ank, Ank_2, Ank_3, Ank_4, Ank_5, PARP	
TERTloc1	JAIFHF010000795.1	1195.7	1724	RVT_1, Telomerase_RBD	YES
TERTloc2	JAIFHF010000565.1	809.1	1280	Ank, Ank_2, Ank_3, Ank_4, Ank_5	
TERTloc3	JAIFHF010005449.1	711.4	1983	Ank, Ank_2, Ank_3, Ank_4, Ank_5, Ank_KRIT1	
TERTloc4	JAIFHF010000117.1	663.7	1993	Ank, Ank_2, Ank_3, Ank_4, Ank_5	
TERTloc5	JAIFHF010000819.1	629.0	1832	Ank, Ank_2, Ank_3, Ank_4, Ank_5	
TERTloc6	JAIFHF010039359.1	598.9	2153	Ank, Ank_2, Ank_3, Ank_4, Ank_5, Ank_KRIT1	
TERTloc7	JAIFHF010004461.1	577.6	1180	Ank, Ank_2, Ank_3, Ank_4, Ank_5, Bactofilin	
TERTloc8	JAIFHF010000263.1	570.6	1108	Ank, Ank_2, Ank_3, Ank_4, Ank_5, Ank_KRIT1	
TERTloc9	JAIFHF010002025.1	554.6	1730	Ank, Ank_2, Ank_3, Ank_4, Ank_5, Ank_KRIT1	
TERTloc10	JAIFHF010016703.1	550.1	1637	Ank, Ank_2, Ank_4, Ank_5	

TERTloc11	JAIFHF010002962.1	525.5	1204	Ank, Ank_2, Ank_3, Ank_4, Ank_5, Ank_KRIT1	
TERTloc12	JAIFHF010017659.1	517.3	1340	Ank, Ank_2, Ank_3, Ank_4, Ank_5	
TERTloc13	JAIFHF010000794.1	514.5	1156	Ank, Ank_2, Ank_3, Ank_4, Ank_5, Ank_KRIT1	
TERTloc14	JAIFHF010011520.1	489.7	1682	Ank, Ank_2, Ank_3, Ank_4, Ank_5, Ank_KRIT1	
TERTloc15	JAIFHF010036037.1	475.9	1177	Ank, Ank_2, Ank_3, Ank_4, Ank_5, Ank_KRIT1	
TERTloc16	JAIFHF010025068.1	445.2	1683	Ank, Ank_2, Ank_3, Ank_4, Ank_5	
TERTloc17	JAIFHF010009054.1	428.8	1830	Ank, Ank_2, Ank_3, Ank_4, Ank_5, Ank_KRIT1	
TERTloc18	JAIFHF010029131.1	381.4	823	Ank, Ank_2, Ank_4, Ank_5	
TERTloc19	JAIFHF010002442.1	377.6	1022	Ank, Ank_2, Ank_3, Ank_4, Ank_5, Ank_KRIT1	
TERTloc20	JAIFHF010040247.1	304.4	419	Ank, Ank_2, Ank_3, Ank_4, Ank_5	

G.Aaurita

Locus	Contig	Bits	aa	Pfam families	Telomerase?
TERTloc0	REGM01000008.1	2038.6	1124	Ank, Ank_2, Ank_3, Ank_4, Ank_5, PARP	
TERTloc1	REGM01000190.1	954.4	1101	Ank, Ank_2, Ank_3, Ank_4, Ank_5, Ank_KRIT1	
TERTloc2	REGM01000013.1	661.5	1108	RVT_1, Telomerase_RBD	YES

TERTloc3	REGM01000015.1	634.0	594	Ank, Ank_2, Ank_3, Ank_4, Ank_5, Ank_KRIT1	
TERTloc4	REGM01000003.1	563.7	949	Ank, Ank_2, Ank_3, Ank_4, Ank_5	
TERTloc5	REGM01000001.1	377.1	540	Ank, Ank_2, Ank_3, Ank_4, Ank_5, Ank_KRIT1	
TERTloc6	REGM01000003.1	322.4	976	ANK_LRRK2, Ank, Ank_2, Ank_4, Ank_KRIT1	
TERTloc7	REGM01000019.1	294.9	453	Ank, Ank_2, Ank_3, Ank_4, Ank_5	
TERTloc8	REGM01000093.1	294.3	517	Ank, Ank_2, Ank_3, Ank_4, Ank_5, Ank_KRIT1	
TERTloc9	REGM01000001.1	292.4	342	Ank, Ank_2, Ank_3, Ank_4, Ank_5, Ank_KRIT1	
TERTloc10	REGM01000245.1	289.6	330	Ank, Ank_2, Ank_3, Ank_4, Ank_5	
TERTloc11	REGM01000019.1	287.3	453	Ank, Ank_2, Ank_3, Ank_4, Ank_5	
TERTloc12	REGM01000040.1	235.7	236	Ank, Ank_2, Ank_3, Ank_4, Ank_5	
TERTloc13	REGM01000101.1	220.7	388	Ank, Ank_2, Ank_3, Ank_4, Ank_5, Ank_KRIT1	
TERTloc14	REGM01000019.1	188.6	203	Ank, Ank_2, Ank_3, Ank_4, Ank_5, T4SS_AnkD	
TERTloc15	REGM01000033.1	181.4	323	Ank, Ank_2, Ank_3, Ank_4, Ank_5	
TERTloc16	REGM01000002.1	127.1	203	Ank, Ank_2, Ank_3, Ank_4, Ank_5	

TERTloc17	REGM01000006.1	120.2	337	Ank, Ank_2, Ank_3, Ank_4, Ank_5	
TERTloc18	REGM01000114.1	119.4	190	ANK_LRRK2, Ank, Ank_2, Ank_3, Ank_4, Ank_5	
TERTloc19	REGM01002319.1	112.5	301	Ank, Ank_2, Ank_3, Ank_4, Ank_5	

Appendix H. Unique substitutions, full lists

H.DKC1 in Tdohrnii (21 substitutions)

Pos	Query	Consensus	Refs	Agree
1	N	M	3	3
2	Y	A	3	3
22	L	K	3	3
24	R	K	3	3
62	A	K	3	3
339	G	N	3	3
484	A	T	3	3
486	E	H	3	3
487	M	D	3	3
488	L	V	3	3
490	G	D	3	3
492	L	Q	3	3
636	E	N	3	3
711	E	D	3	3
727	L	T	3	3
1138	W	K	3	3
1140	S	K	3	3
1141	F	K	3	3
1143	N	K	3	3
1145	R	K	3	3
1151	D	E	3	3

H.DKC1 in Aaurita (5 substitutions)

Pos	Query	Consensus	Refs	Agree
1	L	M	3	3

324	G	A	3	3
839	R	K	3	3
840	R	K	3	3
842	R	K	3	3

H.LIG4 in Tdohrnii (45 substitutions)

Pos	Query	Consensus	Refs	Agree
1	R	M	4	3
7	L	V	4	4
26	D	E	4	3
43	Y	H	4	4
92	F	L	4	3
103	S	A	4	3
219	V	L	3	3
221	K	N	4	3
222	N	Y	4	4
365	I	V	4	3
378	D	E	4	3
382	T	D	4	4
388	L	I	4	3
408	I	L	4	3
804	L	F	3	3
884	V	I	3	3
932	L	M	4	3
950	C	L	4	3
962	Q	E	4	3
1022	T	E	4	3
1025	V	I	4	4
1034	M	K	4	3
1041	I	V	4	3
1072	E	D	3	3
1091	V	M	3	3
1395	V	I	4	3
1398	T	A	4	3
1402	V	P	4	3
1404	E	D	4	3
1425	D	S	4	3
1428	D	E	4	3

1431	S	T	4	3
1436	L	E	4	3
1442	C	A	4	3
1450	I	L	4	3
1452	P	S	4	3
1608	L	F	3	3
1612	Q	E	4	3
1627	F	L	3	3
1729	F	Y	4	3

H.LIG4 in Aaurita (8 substitutions)

Pos	Query	Consensus	Refs	Agree
2	M	I	4	4
12	M	I	4	3
58	V	W	4	3
60	L	I	3	3
61	C	R	3	3
65	Q	K	4	4
92	S	N	4	3
103	D	K	4	3

H.MLH1 in Tdohrnii (21 substitutions)

Pos	Query	Consensus	Refs	Agree
1	L	M	3	3
2	V	A	3	3
16	I	V	4	3
35	L	M	4	3
114	V	T	3	3
118	R	K	4	3
141	N	G	3	3
189	I	V	4	3
192	V	I	4	3
265	Y	L	3	3
266	F	L	3	3
274	Y	I	3	3
301	R	H	4	3
308	I	L	3	3

319	M	V	3	3
333	L	A	4	3
340	R	K	3	3
435	E	Q	4	3
583	R	K	3	3
948	Q	E	3	3
1162	K	R	4	3

H.MLH1 in Aaurita (35 substitutions)

Pos	Query	Consensus	Refs	Agree
10	Q	K	4	3
26	H	Q	4	3
40	I	L	4	3
43	G	K	4	3
79	K	R	4	3
83	L	M	4	3
88	K	E	4	3
96	E	S	4	3
120	I	V	4	4
132	G	S	4	3
144	N	K	4	3
146	I	V	3	3
161	T	K	3	3
167	A	V	3	3
205	I	V	4	3
266	F	Y	4	3
279	N	H	3	3
336	L	A	4	3
341	I	V	4	3
344	A	C	4	3
346	D	E	4	3
748	V	G	4	3
757	I	K	3	3
812	D	E	4	3
821	E	P	4	3
841	E	D	4	3
874	F	Y	4	3
891	D	E	4	4

924	L	F	3	3
927	V	E	4	3
933	A	S	4	3
1022	Q	E	3	3
1039	L	F	4	3
1043	C	G	4	3
1047	Q	E	4	3

H.MSH2 in Tdohrnii (79 substitutions)

Pos	Query	Consensus	Refs	Agree
1	V	M	4	3
47	Y	F	4	3
48	F	Y	4	3
59	R	A	3	3
61	L	E	4	3
63	K	F	4	3
64	F	K	4	3
65	C	T	4	4
67	C	S	4	3
68	L	V	4	4
78	S	K	4	3
160	F	Y	4	4
161	S	R	4	3
219	A	V	4	3
234	I	L	4	4
236	T	G	4	3
238	T	N	4	3
395	N	D	4	4
406	Y	F	4	4
415	T	L	4	3
502	R	N	4	3
514	Y	F	4	3
824	D	E	4	3
830	Y	F	4	3
832	I	L	4	3
835	L	F	4	4
867	T	S	4	4
869	C	S	4	3

887	G	A	4	3
890	V	I	4	3
978	Q	K	3	3
979	V	I	4	3
980	D	E	4	4
985	M	I	4	3
988	V	A	4	4
992	N	D	4	3
994	D	E	4	3
995	I	L	4	3
998	S	T	4	3
999	I	L	4	3

H.MSH2 in Aaurita (65 substitutions)

Pos	Query	Consensus	Refs	Agree
90	S	D	4	3
91	K	R	4	3
113	R	K	3	3
134	Y	S	3	3
135	A	V	3	3
166	R	K	4	4
173	S	V	4	3
186	E	D	4	3
241	I	V	4	4
265	E	D	4	3
313	V	I	3	3
492	V	I	3	3
493	R	K	3	3
505	Q	E	4	4
518	I	L	4	3
556	C	S	4	3
577	V	I	4	3
588	G	E	4	3
589	K	E	4	3
638	I	V	4	3
644	R	Q	4	3
659	N	H	4	3
690	D	E	3	3

702	D	E	3	3
705	D	E	3	3
818	K	N	3	3
822	D	G	3	3
834	K	R	4	4
836	S	T	4	4
838	Q	K	4	4
845	Q	N	4	3
869	M	L	4	4
881	T	N	4	3
882	S	E	4	3
897	N	K	4	3
902	L	I	4	4
918	V	I	4	3
929	R	K	4	3
930	G	D	4	3
938	A	T	4	3

Appendix I. ML seed-level metrics

I.1 CNN within-species and transfer

Setting	CNN mean	CNN sd	k-mer LR mean	LR sd
within Tdohnii	0.620	0.066	0.646	0.075
within Aurita	0.515	0.113	0.657	0.088
Tdohnii -> Aurita	0.475	0.046	0.470	0.000
Aurita -> Tdohnii	0.480	0.019	0.495	0.000
GNN species discrim.	0.487	0.098	0.860	0.063

Appendix J. SRA run census detail

J.Turritopsis dohrnii: 60 runs

Library strategy	Runs
RNA-Seq	58
WGS	2

J.Aurelia aurita: 206 runs

Library strategy	Runs
RNA-Seq	100

WGS	49
AMPLICON	40
WGA	15
EST	2

Appendix K. TERC records used

Accession	Organism	Role
BK012923.1	<i>Actinia tenebrosa</i>	query
BK012924.1	<i>Corynactis australis</i>	query
BK012925.1	<i>Ricordea yuma</i>	query
BK012926.1	<i>Aurelia aurita</i>	query + positive control
BK012927.1	<i>Chrysaora fuscescens</i>	query
BK012928.1	<i>Rhopilema esculentum</i>	query
BK012929.1	<i>Periphylla periphylla</i>	query
BK012930.1	<i>Calvadosia cruxmelitensis</i>	query
BK012931.1	<i>Lucernaria quadricornis</i>	query
BK012932.1	<i>Craspedacusta sowerbii</i>	query

Appendix L. Genome assembly statistics

Assembly	Sequences	Size (bp)	End-analysis seqs
Tdohrnii_Kazusa_GCA_027922465.2	891	435923997	882
Tdohrnii_Oviedo_GCA_025167195.1	74835	400131687	1214
Aaurita_GCA_004194415.1	2709	376932402	739
Clytia_GCF_902728285.1	1396	420978673	772
HydraT2T_GCF_038396675.1	15	911675201	15
Nematostella_GCA_932526225.2	54	269889592	36

Appendix M. Pooled upstream motif leaders (both jellies)

T. dohrnii 6-mer	z	A. aurita 6-mer	z
CTTAGC	5.88	ACTCGG	13.68
CTAAGA	3.81	ACCGAG	10.25
AAATAA	3.54	AACCGA	9.13
GCACGC	3.44	AAACCG	8.43
AAACTT	3.42	AACTCG	7.83
AGGATG	3.19	AACCCG	6.64
CCATCC	3.18	AAACTC	5.76

ATTACA	3.1	AGATAA	5.56
AAGGCT	3.06	ATAAAC	5.49
CGCACG	3.02	AGCCGG	5.25
AATAAA	2.97	ACCCGG	5.06
CTCAAC	2.93	AAACTA	4.73
GCCACA	2.93	ACAGTA	4.59
GACAAC	2.89	AAAACC	4.54
TACAAA	2.88	CAGATA	4.53
TGCTAA	2.88	ACTATA	4.45
CACACA	2.86	AAACCC	4.45
ATCCTG	2.8	AGTACT	4.4
AAAACCT	2.75	CTCCAG	4.32
CACGCC	2.75	CAGTAC	4.28